Tidyverse and ggplot Review

1 Overview

This is a review of important concepts in tidyverse and ggplot that can be used to summarize data. It covers three main sections: data cleaning, summarizing categorical data, and summarizing continuous data, followed by two real-world examples of how these tools can be applied.

2 Data Cleaning

2.1 Summarizing important Operator in Tidyverse

When we want to analyze data, the real-world data is often messy and unstructured. We can use operators in the tidyverse to clean and filter the data to extract what we need, and at the same time compute the statistics we are interested in.

`tibble()`, constructs a tibble, an enhanced data frame

`read_csv()`, like read.csv, but quicker and returns a tibble

`mutate()`, used to create new variables or edit old variables

`filter()`, select rows that satisfy provided condition(s)

`group_by()`, for specifying a group variable for data operations

`|>` pipes, takes the result on the left and passes it as the first argument to the function on the right.

`select()`, select one or more columns as a data frame

`case_when()`, create a new variable by assigning different values based on multiple conditions.

`summarise()`, allows us to ask for statistics calculated on selected column(s). Those statistics includes:

- `mean()`
- `sd()`
- `min()`
- `quantile()`
- `median()`
- `max()`
- `IQR()`
- `e1071::skewness()`
- `e1071::kurtosis()`

3 Categorical Data

Categorical variable places observations into one of several groups or categories.

- Nominal: assumes groups are made up of named categories.

- Ordinal: assumes groups are made up of ordered categories

3.1 Numerical Summary

When summarizing data, We want to compute the sample proportion of observations that belong to each level of the categorical variable.

$$\hat{p} = \frac{x}{n}$$

- x is the count of observations in the specified level
- n is the count of observations in total

We can use `group_by()` to group the data into the level we need.

3.2 Graphical Summary

One Categorical Variable:

We can use a bar plot, pie chart, or doughnut plot to visualize the data. It is important to note that we need to set the y-axis to proportions instead of counts, because we want to understand each category relative to the overall population rather than just the raw frequency.

- Bar plot: `geom_bar()`
- Pie chart: `geom_bar()` + `coord_polar("y", start=0)`
- Doughnut Plot: `geom_bar()` + `coord_polar("y", start=0)` + `xlim(0.5, 2.5)`

Two Categorical Variable:

We are usually interested in understanding how the distribution of one variable changes across the levels of another variable. This is typically done by creating a contingency table, where we count the number of observations for each combination of categories. Graphically we can create grouped bar plot or grouped pie chart.

- Grouped bar plot: `geom_bar()` + `facet_wrap()` or `geom_bar(...fill = )`
- Grouped pie chart: `geom_bar()` + `coord_polar("y", start=0)` + `facet_wrap()`

4 Example of Summarizing Categorical and Continuous Data

Brockmann (1996) collected data on the mating behavior of nesting female horseshoe crabs, and their research question is what attributes of a female horseshoe crab lead to having more satellites.

If we want to know the distribution of crab color in the data, we could create a summary table and a bar plot.

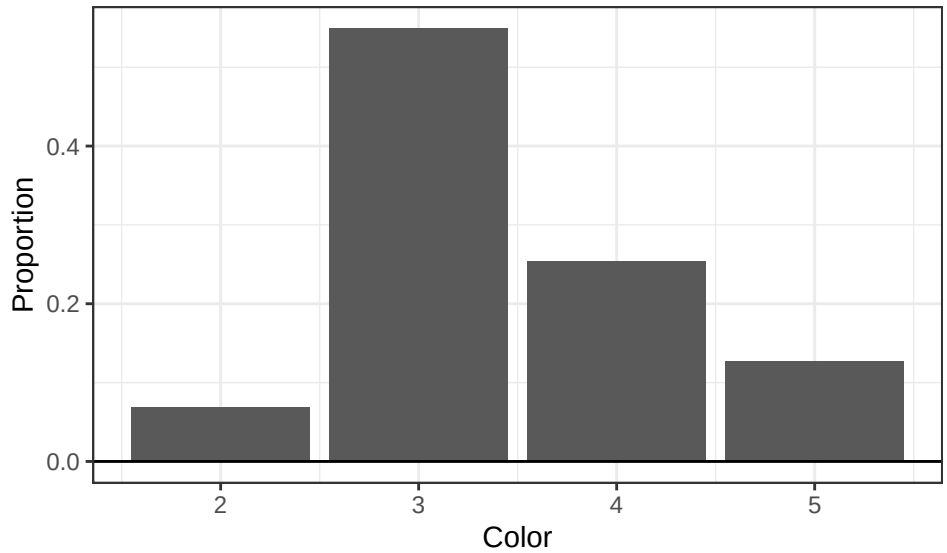
```r
data <- read.csv('Brockmann96.csv')

data.summary <- data |>
  count(color) |>
  mutate(Proportion = n/sum(n))
```

Table 1: Colors of crabs in the data

color	n	Proportion
2	12	0.07
3	95	0.55
4	44	0.25
5	22	0.13

```r
ggplot(data) +
  geom_bar(aes(x = color,
               y = after_stat(count/sum(count)))) +
  geom_hline(yintercept = 0)+
  theme_bw() +
  labs(x = "Color",
       y = "Proportion")
```

Figure 1: Colors of crabs in the data



We can also summarize data of the spine condition of each color of the crabs.

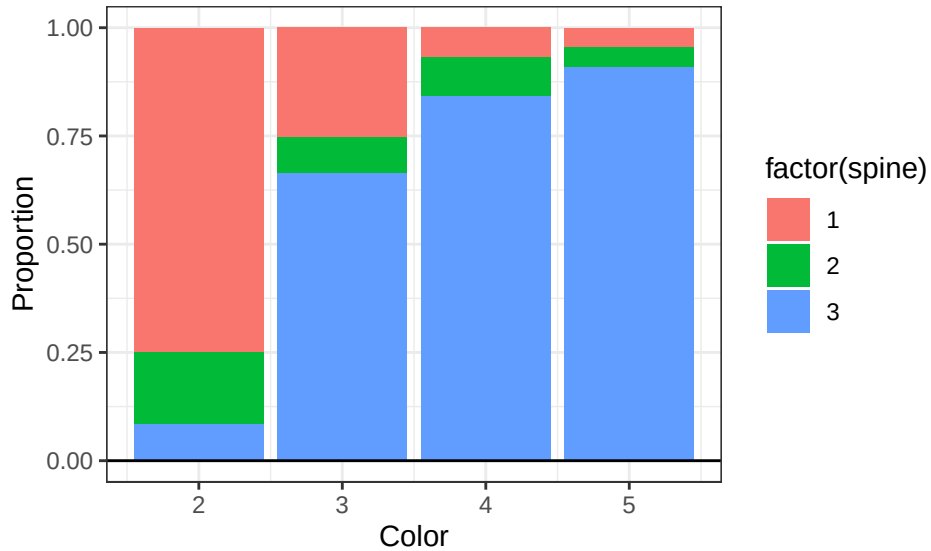
```r
data.summary <- data |>
  count(color, spine) |>
  mutate(prop = n / sum(n), .by = color)
```

Table 2: Colors and spine conditions of crabs in the data

color	spine	n	prop
2	1	9	0.75
2	2	2	0.17
2	3	1	0.08
3	1	24	0.25
3	2	8	0.08
3	3	63	0.66
4	1	3	0.07
4	2	4	0.09
4	3	37	0.84
5	1	1	0.05
5	2	1	0.05
5	3	20	0.91

```r
ggplot(data.summary) +
  geom_bar(aes(x = color,
               y = prop, fill = factor(spine)),
               stat="identity") +
  geom_hline(yintercept = 0)+
  theme_bw() +
  labs(x = "Color",
       y = "Proportion")
```

Figure 2: Colors and spine conditions of crabs in the data



To answer the research question, we can create a grouped box plot and comparing how does difference in color contributes to the difference in number of satellites.

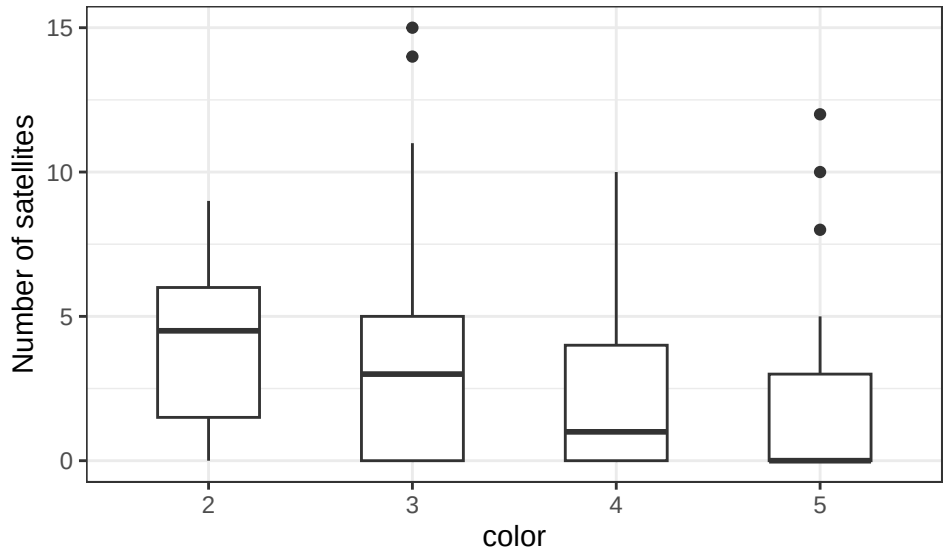
```
data.summary <- data|>
  group_by(color) |>
  summarize(
    Mean = mean(num.satellites),
    SD = sd(num.satellites),
    Median = median(num.satellites),
    IQR = IQR(num.satellites),
    Skewness = e1071::skewness(num.satellites),
    "Excess Kurtosis" = e1071::kurtosis(num.satellites)
)
```

Table 3: Colors and number of satellites of crabs

color	Mean	SD	Median	IQR	Skewness	Excess Kurtosis
2	4.08	3.12	4.5	4.5	-0.04	-1.49
3	3.29	3.21	3.0	5.0	1.11	1.48
4	2.23	2.60	1.0	4.0	1.12	0.66
5	2.05	3.62	0.0	3.0	1.54	1.09

```
ggplot(data) +
  geom_boxplot(aes(x = factor(color),
                   y = num.satellites), width = 0.5)+
  theme_bw() +
  labs(x = "color",
       y = "Number of satellites")
```

Figure 3: Colors and number of satellites of crabs



Color number 1-5 represent from light to dark, and the statistic and graph seem to suggests that crabs with lighter color tend to have higher number of satellites, although there are also some outliers in darker color crabs.

5 Quantitative Data

Quantitative data refers to numerical variables that represent measurable quantities.

- Continuous: assumes the variable is double measured
- Discrete: assumes the variable is countable

5.1 Numerical Summary

When summarizing numerical data, we are interested in describing its center, such as the mean and median, its variability, such as the standard deviation, and its distributional shape, such as skewness and kurtosis.

- Mean:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- Sample Variance:

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

- Sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

- Median:
  - The median is the observation in the middle of the data when placed in ascending order.
- Interquatile Range:
  - The first quartile Q_1 is the median of the lower half of the observations, making it the 25th percentile.

  - The third quartile Q_3 is the median of the upper half, making it the 75th percentile.

$$IQR = Q_3 - Q_1$$

- Skewness
  - When skewness < 0, there are more observations for high values and fewer with low values.
  - When skewness $= 0$, the data are symmetric – this means that the data are symmetrically distributed around the mean.
  - When skewness > 0, there are more observations for low values and fewer with high values.
- Excess Kurtosis

  The excess kurtosis describes how peaked data are in relation to the Gaussian distribution (the bell-shaped curve) with the same mean and variance.
  - When excess kurtosis < 0, the data are platykurtic – the data have a flatter peak and exhibit more variability.
  - When excess kurtosis $= 0$, the data are mesokurtic – the data have a peak and tails similar to the Gaussian distribution.
  - When excess kurtosis > 0, the data are leptokurtic – the data have a more pronounced peak and exhibit less variability.
- Correlation

  We can use `cor(... method = '')` to measure the correlation between two continuous variables.
  - Pearson Correlation Coefficient (r)
  - Spearman's Rank Correlation ( )
  - Kendall's Tau ( )

5.2 Graphical Summary

One continuous variable

If we are interested in the distributiton shape, we can create a histogram or a violin plot. If we are interested in key statistics such as the median, quartiles, and potential outliers, we can create a box plot. We can also create a box plot and violin plot together if we want to know both.

- Histogram: `geom_histogram()`
- Violin Plot: `geom_violin()`
- Box Plot: `geom_boxplot()`

Two continuous variable

We are generally interested in the correlation between the two variable, so we can create a scatter plot to see whether they move in the same direction.

- Scatter Plot: `geom_point()` + `geom_smooth()`

6 Example of Data Cleaning and Summarizing Quantitative Data

In my econometric class, I worked on a research project that studied how does the changes in petroleum price affects PM2.5 levels in the air. I used R to conduct lots of data cleaning and summarizing process, and I think those data serve as a good example of reviewing concepts about tidyverse and ggplot.

PM2.5 refers to airborne particulate matter with a diameter of 2.5 micrometers or smaller. These fine particles can penetrate deep into the lungs and even enter the bloodstream, posing significant risks to respiratory and cardiovascular health.

In theory, a higher petroleum price will decrease consumption and thus decrease pollutants such as PM2.5 in the air. However, whether this causal relationship applies to reality and the extent of this impact remains uncertain.

I obtain data of PM2.5 level from the United States Environmental Protection Agency (U.S. Environmental Protection Agency 2026) and petroleum price data from U.S. Energy Information Administration (U.S. Energy Information Administration 2026)

6.1 Loading the Data

We can use `read.csv()` to read csv.files

For example, I have data of average air pollutants in the air by county in 2000 from the EPA(cite).

```
dat.air <- read.csv(file = 'EPA2000.csv')
```

6.2 Selecting columns

We can use `data$column` to select column of the data.

```
state <- dat.air$State.Name
```

If we are only need information about the county code, state name, parameter name, and arithmetic mean, we can use `select()` to select multiple columns. We can use `head()` to check the head of the data

```
dat.air.clean <- dat.air |> select('County.Code', 'State.Name', 'Parameter.Name', 'Arithmetic.Mean')
head(dat.air.clean)
```

```
  County.Code State.Name       Parameter.Name Arithmetic.Mean
1           3    Alabama                Ozone        0.061870
2           3    Alabama                Ozone        0.054953
3           3    Alabama                Ozone        0.054953
4           3    Alabama                Ozone        0.054835
5           3    Alabama Sample Flow Rate- CV        0.288660
6           3    Alabama        Sample Volume       23.985567
```

6.3 Filtering Data

Since we only want the data of PM2.5, we can use `filter()` to exclude any other parameter.

```
dat.air.clean <- dat.air.clean |> filter(Parameter.Name == 'PM2.5 - Local Conditions')
```

We also want to only include observations within the U.S.

```
dat.air.clean <- dat.air.clean |> filter(!State.Name %in% c(
    "Canada",
    "Country Of Mexico",
    "Puerto Rico",
    "Virgin Islands"
  ))
```

We can also use `filter()` to drop NAs. `is.na()` will return whether the value the NA.

```
dat.air.clean <- dat.air.clean |> filter(!is.na(Arithmetic.Mean))
head(dat.air.clean)
```

```
  County.Code State.Name           Parameter.Name Arithmetic.Mean
1           3    Alabama PM2.5 - Local Conditions        14.83412
2           3    Alabama PM2.5 - Local Conditions        14.83412
3           3    Alabama PM2.5 - Local Conditions        14.83412
4           3    Alabama PM2.5 - Local Conditions        14.83412
5           3    Alabama PM2.5 - Local Conditions        14.83412
6           3    Alabama PM2.5 - Local Conditions        14.83412
```

6.4 Mutate

Each county has lots of different measurement of the arithmetic mean, and I want to add a new column that contain the average level of PM2.5 of that county. We can use `group_by()` and `mutate()` to accomplish this task. If the data still contain empty value, we need to set `na.rm = TRUE` so that we do not get empty values.

```
dat.air.clean <- dat.air.clean |>
    mutate(pm25_avg = mean(Arithmetic.Mean, na.rm = TRUE),
           .by = County.Code)
head(dat.air.clean)
```

```
  County.Code State.Name           Parameter.Name Arithmetic.Mean pm25_avg
1           3    Alabama PM2.5 - Local Conditions        14.83412 11.93812
2           3    Alabama PM2.5 - Local Conditions        14.83412 11.93812
3           3    Alabama PM2.5 - Local Conditions        14.83412 11.93812
4           3    Alabama PM2.5 - Local Conditions        14.83412 11.93812
5           3    Alabama PM2.5 - Local Conditions        14.83412 11.93812
6           3    Alabama PM2.5 - Local Conditions        14.83412 11.93812
```

6.5 Numerical Summary

We can use `summarize()` to create a summary table. Here I will use the fully clean version of data that contain information of year, petroleum price, PM2.5 level, state name, and county code.

```
data.total <- read.csv(file = 'totaldata.csv')

head(data.total)
```

```
  year Petroleum.Price State.Name County.Code     PM2.5 Natural.Gas.Price
1 2000            8.54     Alaska          20  5.670303              1.97
2 2000            8.54     Alaska          68  1.619042              1.97
3 2000            8.54     Alaska          90 14.447360              1.97
4 2000            8.54     Alaska         110  5.926137              1.97
5 2000            8.54     Alaska         130  5.885000              1.97
6 2000            8.54     Alaska         170  4.547621              1.97
  Coal.Price  log_pm25 log_ptroleum
1       1.87 1.7352426     2.144761
2       1.87 0.4818348     2.144761
3       1.87 2.6705117     2.144761
4       1.87 1.7793726     2.144761
5       1.87 1.7724067     2.144761
6       1.87 1.5146043     2.144761
```

```
data.summary <- data.total |>
  drop_na(PM2.5) |>
  summarize(
    Mean = mean(PM2.5),
    SD = sd(PM2.5),
    Median = median(PM2.5),
    IQR = IQR(PM2.5),
    Skewness = e1071::skewness(PM2.5),
    "Excess Kurtosis" = e1071::kurtosis(PM2.5)
  )
```

Table 4: Summary of PM2.5 levels

Mean	SD	Median	IQR	Skewness	Excess Kurtosis
9.81	2.49	9.49	3.46	0.64	4.03

Since the data contain information of PM2.5 in each year, we can create summary table that is grouped by year.

```r
data.summary <- data.total |>
  filter(year <= 2003) |>       # I only include 4 years in the summary
  drop_na(PM2.5) |>
  group_by(year) |>
  summarize(
    Mean = mean(PM2.5),
    SD = sd(PM2.5),
    Median = median(PM2.5),
    IQR = IQR(PM2.5),
    Skewness = e1071::skewness(PM2.5),
    "Excess Kurtosis" = e1071::kurtosis(PM2.5)
  )
```

Table 5: Summary of PM2.5 levels2

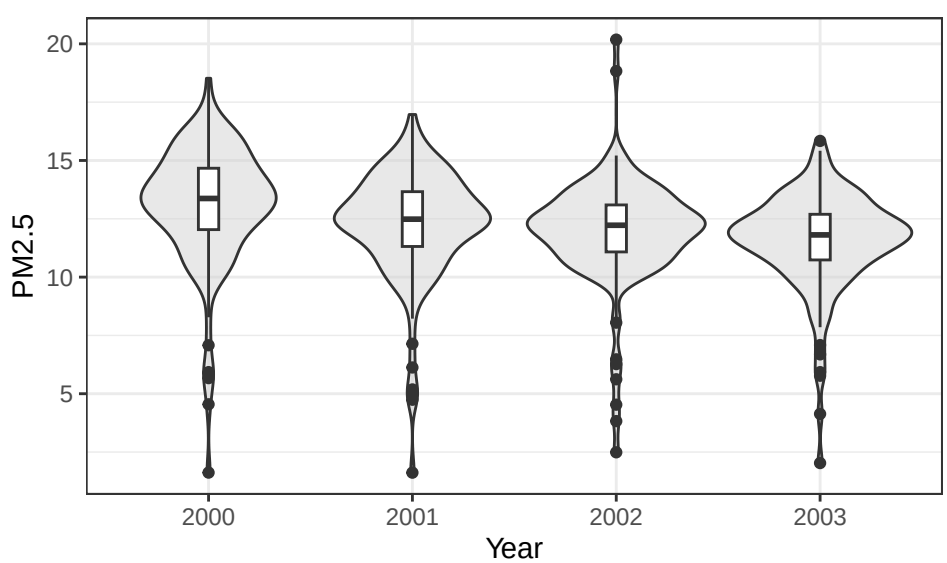
year	Mean	SD	Median	IQR	Skewness	Excess Kurtosis
2000	13.10	2.56	13.37	2.63	-1.39	3.55
2001	12.27	2.32	12.48	2.35	-1.47	3.94
2002	11.96	2.25	12.22	2.01	-0.94	4.98
2003	11.55	2.06	11.81	1.96	-1.42	3.97

6.6 Graphical Summary

We can create box plot or violin plot to graphical illustrates the distribution of average PM2.5 in each year.

```r
data.new <- data.total |>
  filter(year <= 2003)

ggplot(data.new) +
  geom_violin(aes(x = factor(year), y = PM2.5),
              fill = "lightgrey", alpha=0.5) +
  geom_boxplot(aes(x = factor(year), y = PM2.5),width = 0.1)+
  theme_bw() +
  labs(x = "Year", y = "PM2.5")
```

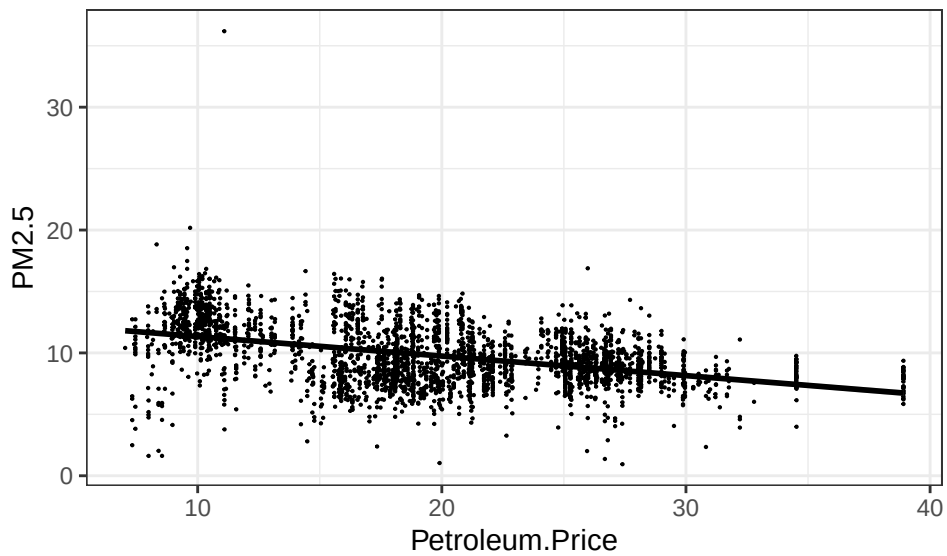
Figure 4: Yearly PM2.5 Concentration Distribution



We can create a scatter plot to see if there is a correlation between petroleum price and PM2.5 level.

```r
ggplot(data.total) +
  geom_point(aes(x = Petroleum.Price, y = PM2.5), size = 0.1) +
  geom_smooth(aes(x = Petroleum.Price, y = PM2.5),
                  method = "lm", color = "black") +
  theme_bw()
```

Figure 5: Petroleum Price and PM2.5 level in the air



The estimate coefficient is biased here because I did not add any control variable and time state fixed effect in this model. However, this still serve as an example of how we summary two continuous variable data.

```r
data.summary <- data.total |>
  drop_na(Petroleum.Price, PM2.5) |>
  summarize(
    Pearson = cor(Petroleum.Price, PM2.5, method="pearson"),
    Spearman = cor(Petroleum.Price, PM2.5, method="spearman"),
    Kendall = cor(Petroleum.Price, PM2.5, method="kendall")
)
```

Table 6: Correlation between Petroleum Price and PM2.5 levels

Pearson	Spearman	Kendall
-0.43	-0.41	-0.28

References

Brockmann, H Jane. 1996. "Satellite Male Groups in Horseshoe Crabs, Limulus Polyphemus." *Ethology* 102 (1): 1–21.

U.S. Energy Information Administration. 2026. "State Energy Data System (SEDS): Prices and Expenditures Data." https://www.eia.gov/state/seds/seds-data-complete.php#PricesExpenditures.

U.S. Environmental Protection Agency. 2026. "AirData Download Files." https://aqs.epa.gov/aqsweb/airdata/download_files.html.